

OUTPUT (695x983)

DMV USE ONLY
 DL address
 obtained by FD
 24002

**A SEPARATE FORM IS NEEDED
 FOR EACH DRIVER OR VEHICLE OWNER**

NOTICE OF CHANGE OF ADDRESS

DMV
 A Public Service Agency

MAIL TO:
 DMV CHANGE OF ADDRESS
 P.O. BOX 942555
 SACRAMENTO, CA 94255-0001

PLEASE PRINT CHARACTERS IN CAPITAL LETTERS USING BLACK OR DARK BLUE INK ONLY.

1 2 3 4 5 6 7 8 9 0 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

INSTRUCTIONS:

- Enter the information as shown on the document. (i.e. California Driver License, ID card, or vehicle registration card, for which a change is being requested)
- Names not matching DMV records and/or inconsistent information cannot be updated.
- Type in your new address on a final page of paper with your signature and date. Keep it with your driver license or ID card.
- If commercial licensed driver must maintain a California residential address or the driver license will be downgraded to non-commercial status.

Personal Information

LAST NAME
 MALICK

FIRST
 SATYA

DATE OF BIRTH
 01 01 1990

DATE OF CHANGE OF ADDRESS
 01 23 2007

Voter Change of Address

Use only with:
 DL Change of Address

New or Correct Residence Address

On Not Use P.O. Box
 in this space

Mark this box if you do not want to change your voting address.

STREET NUMBER ONLY
 10101

STREET NAME
 BINARY STREET

APT NO.

CITY - DO NOT ABBREVIATE - USE FIRST 22 CHARACTERS IN CITY NAME
 ARTIFICIAL INTELLIGENCE

STATE
 CA

ZIP CODE
 10101

New or Correct Mailing Address

On Not Use P.O. Box
 in this space

STREET NUMBER ONLY
 256

STREET NAME
 ARTIFICIAL INTELLIGENCE STREET

APT NO.

CITY - DO NOT ABBREVIATE - USE FIRST 22 CHARACTERS IN CITY NAME
 ARTIFICIAL INTELLIGENCE

STATE
 CA

ZIP CODE
 12345

Leased Vehicles

Use Additional Forms if Necessary

LEASING COMPANY'S NAME
 LEARNOPENCV.COM

Location of Trailer Coach or Vessel

If Different From Residence Address

STREET NUMBER

STREET NAME

CITY - DO NOT ABBREVIATE - USE FIRST 18 CHARACTERS IN CITY NAME

STATE

ZIP CODE

COUNTY - DO NOT ABBREVIATE

Old Address

Your mailing address may be given to requesters providing a valid reason for requesting the information. If you receive mail at your residence, then giving DMV a separate mailing address is optional. Your residence address is restricted to authorized requesters per Vehicle Code Section 1808.21. I am the person whose name appears on the record(s) above and the mailing address shown is valid, existing and accurate. I consent to receive service of process at this mailing address pursuant to 415.20(b), 415.30, and 416.90 of the Civil Procedure Code. I certify (or declare) under penalty of perjury under the laws of the State of California that the foregoing is true and correct.

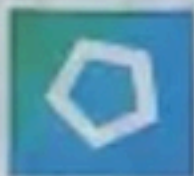
SIGNATURE
 L

DATE
 X

DMV 14 (REV. 02/01) WWW

MUSTER

Finanzierungsplan: 11.11.2012
unter dem Leasingplan: 11.11.2012
Leasingplan: 11.11.2012



MUSTER

Rechnung Nr. 12345

Wir bestätigen und bestätigen die gesamte Zusammenfassung und stellen Ihnen eine Zusammenfassung der folgenden
Lieferanten und Lieferanten in Rechnung.

Nr.	Bezeichnung	Menge	Einheit	Gesamt
1	Fernseher 42 Zoll	1 Stück	1000,00	1000,00
2	Küche und Aufbau	1 Stück	1000,00	1000,00
Summe netto				1.120,00
Umsatzsteuer 19%				212,00
Rechnungsbetrag				1.332,00

Die Rechnung ist somit fertig.

Siehe Übersicht der Rechnungsbeträge. Bitte prüfen Sie die Rechnung auf unser Bankkonto.

Von: 11.11.2012
An: 11.11.2012
E-Mail: 11.11.2012
Web: 11.11.2012

Musterfirma
Vollständige Adresse
11.11.2012
11.11.2012

MUSTER

Finanzierungsplan: 11.11.2012
unter dem Leasingplan: 11.11.2012
Leasingplan: 11.11.2012



MUSTER

Rechnung Nr. 12345

Wir bestätigen und bestätigen die gesamte Zusammenfassung und stellen Ihnen eine Zusammenfassung der folgenden
Lieferanten und Lieferanten in Rechnung.

Nr.	Bezeichnung	Menge	Einheit	Gesamt
1	Fernseher 42 Zoll	1 Stück	1000,00	1000,00
2	Küche und Aufbau	1 Stück	1000,00	1000,00
Summe netto				1.120,00
Umsatzsteuer 19%				212,00
Rechnungsbetrag				1.332,00

Die Rechnung ist somit fertig.

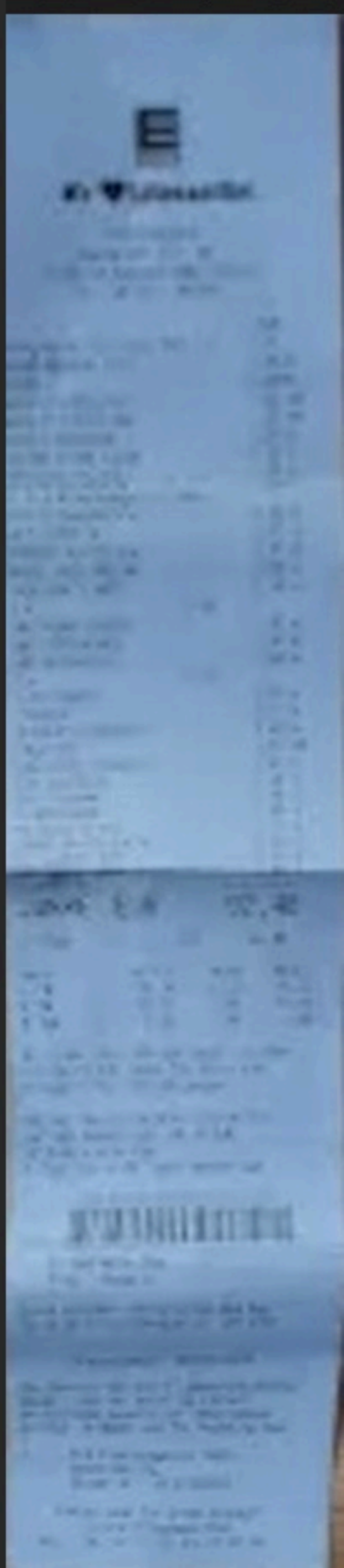
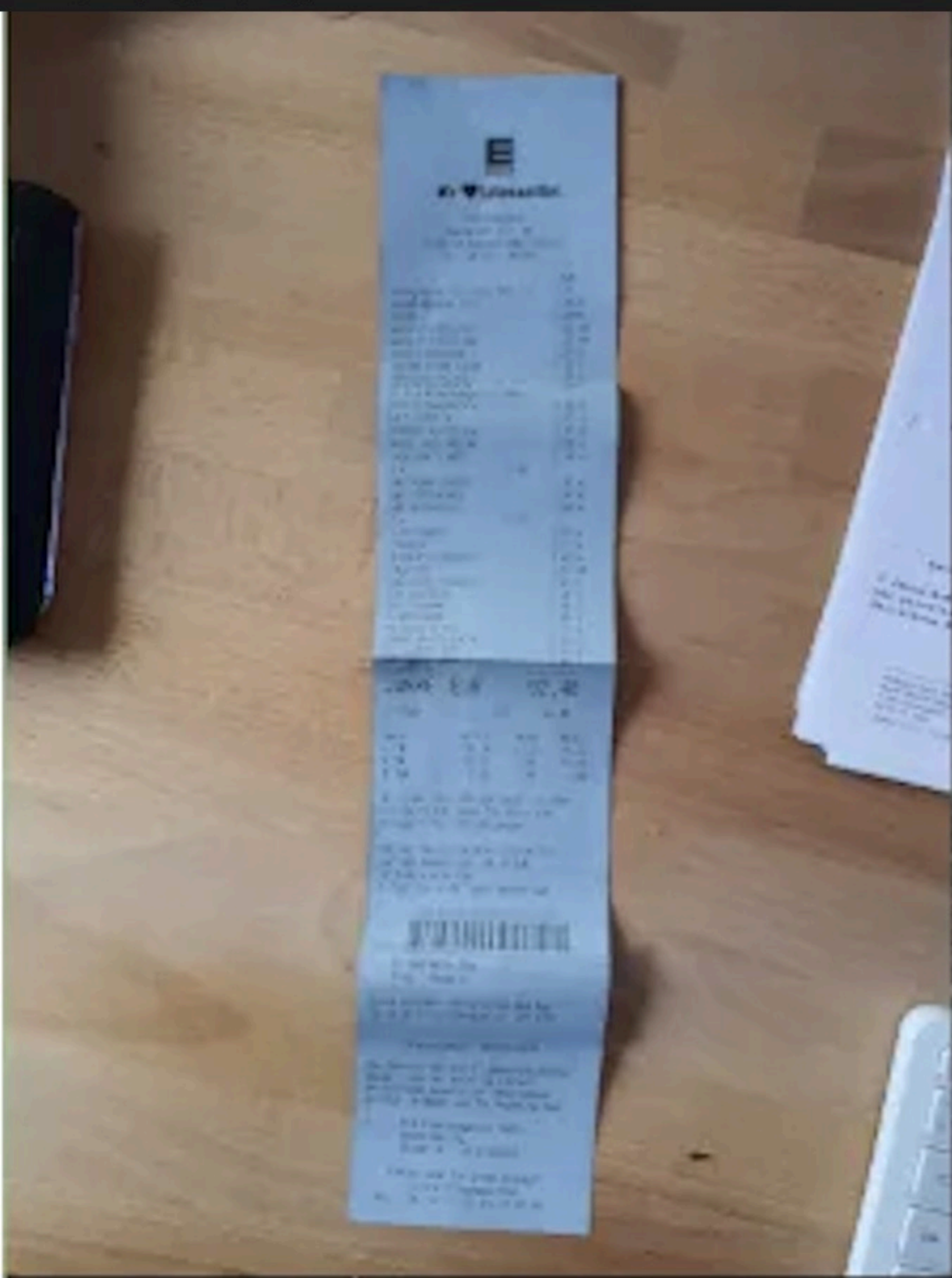
Siehe Übersicht der Rechnungsbeträge. Bitte prüfen Sie die Rechnung auf unser Bankkonto.

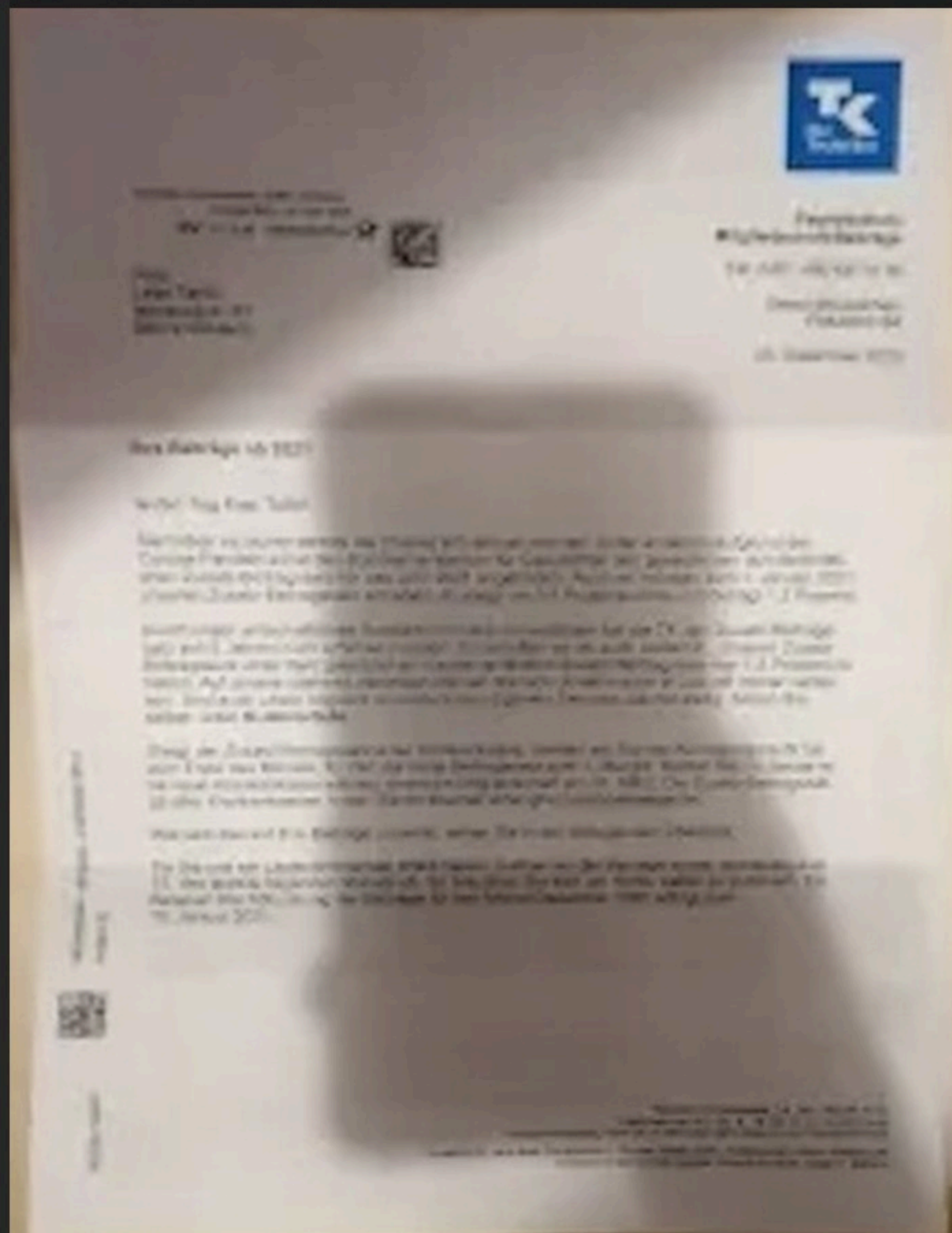
Von: 11.11.2012
An: 11.11.2012
E-Mail: 11.11.2012
Web: 11.11.2012

Musterfirma
Vollständige Adresse
11.11.2012
11.11.2012

INPUT (450x596)

OUTPUT (118x5





These figures reflect a trend in U.S. real property
rights of interest in real property. These figures reflect
the value of real property in the U.S. and the value of
the U.S. real property in the U.S. and the value of the U.S. real property in the U.S.

Only Father thought some better
 But who said the father a better
 It's just it is my father, it will mean my father's better
 But what of father's better will make my father better
 The answer better than Father thought a lot of father is

[illegible]

* I thought I thought but the thought I thought would never thought I thought I thought
in the thought I thought I thought when the thought thought I thought when
thought we were.

Figure 2. *g* and *g*₀ as a function of *g*₀ for *g* = 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0.

Received 10 May 2006; revised 12 July 2006; accepted 12 July 2006
Published online 12 September 2006 in Wiley InterScience (www.interscience.wiley.com). DOI: 10.1002/anie.200601111

These figures represent a portion of the total support. Support of individual supporters. From 1990 to 1994, 17,000 people gave a total of \$1,000,000 to support the project. The bulk of the support came from the state of

Only Father should count water
But who said the count is better
If put it in my water, it will count my Father's love
But what if Father's love will make my Father's love
The love of Father's love will make a lot of love to me

• The main goal of this research is to develop a new method for the detection of the presence of a fault in a system. The method is based on the use of a neural network. The neural network is trained to detect the presence of a fault in a system. The neural network is trained using a set of data. The neural network is trained to detect the presence of a fault in a system. The neural network is trained to detect the presence of a fault in a system.

[illegible]

There are a number of options to help you at the gym, so don't be shy.

© 2004 by The McGraw-Hill Companies, Inc. All rights reserved. Printed in the United States of America. This publication is protected by copyright. Any unauthorized distribution or reproduction of this work is illegal. All other rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording, or by any information storage or retrieval system, without prior written permission from The McGraw-Hill Companies, Inc. This publication contains information that may be confidential or otherwise subject to legal privilege. It is to be used solely for the purpose intended and is not to be distributed outside the intended audience. The McGraw-Hill Companies, Inc. and its subsidiaries and affiliates assume no responsibility for the use or misuse of the information contained herein. The McGraw-Hill Companies, Inc. and its subsidiaries and affiliates assume no responsibility for the accuracy or completeness of the information contained herein. The McGraw-Hill Companies, Inc. and its subsidiaries and affiliates assume no responsibility for the results or consequences of the use of the information contained herein. The McGraw-Hill Companies, Inc. and its subsidiaries and affiliates assume no responsibility for the opinions or conclusions expressed herein. The McGraw-Hill Companies, Inc. and its subsidiaries and affiliates assume no responsibility for the results or consequences of the use of the information contained herein. The McGraw-Hill Companies, Inc. and its subsidiaries and affiliates assume no responsibility for the opinions or conclusions expressed herein.

INPUT (438x582)

SCREWFIXUnit 13, Spa Industrial Park
Longfield Road
Tunbridge Wells, Kent
TN2 3ENOpening Hours
Mon - Fri 7am - 8pm
Sat 7am - 8pm
Sun 9am - 4pmInvoice Number **A4325601005**

Date 21/05/2018

Order DetailsGoods available to collect
From 22-May-2018 after 12 noon

48364	BOSCH ROTAK 320ER ROTARY LAWNMOWER	69.99	VAT 1 (20%)
-------	------------------------------------	-------	-------------

Total 69.99**Payment Details**

Payment Methods	
Credit Card (****6516)	69.99

VAT Breakdown

Rate	Gross	Net	VAT
VAT @ 20%	69.99	58.33	11.66

Collection No: T367

Screwfix Direct Ltd Reg Office: Trade House Mead Avenue Yeovil BA22 8RT
Reg. No. 3006378 Registered in England Vat No: 232 5555 75
Telephone: 03330 112 112 24 hours a day, 7 days a week Online: www.screwfix.com

OUTPUT (337x543)

SCREWFIXUnit 13, Spa Industrial Park
Longfield Road
Tunbridge Wells, Kent
TN2 3ENOpening Hours
Mon - Fri 7am - 8pm
Sat 7am - 8pm
Sun 9am - 4pmInvoice Number **A4325601005**

Date 21/05/2018

Order DetailsGoods available to collect
From 22-May-2018 after 12 noon

48364	BOSCH ROTAK 320ER ROTARY LAWNMOWER	69.99	VAT 1 (20%)
-------	------------------------------------	-------	-------------

Total 69.99**Payment Details**

Payment Methods	
Credit Card (****6516)	69.99

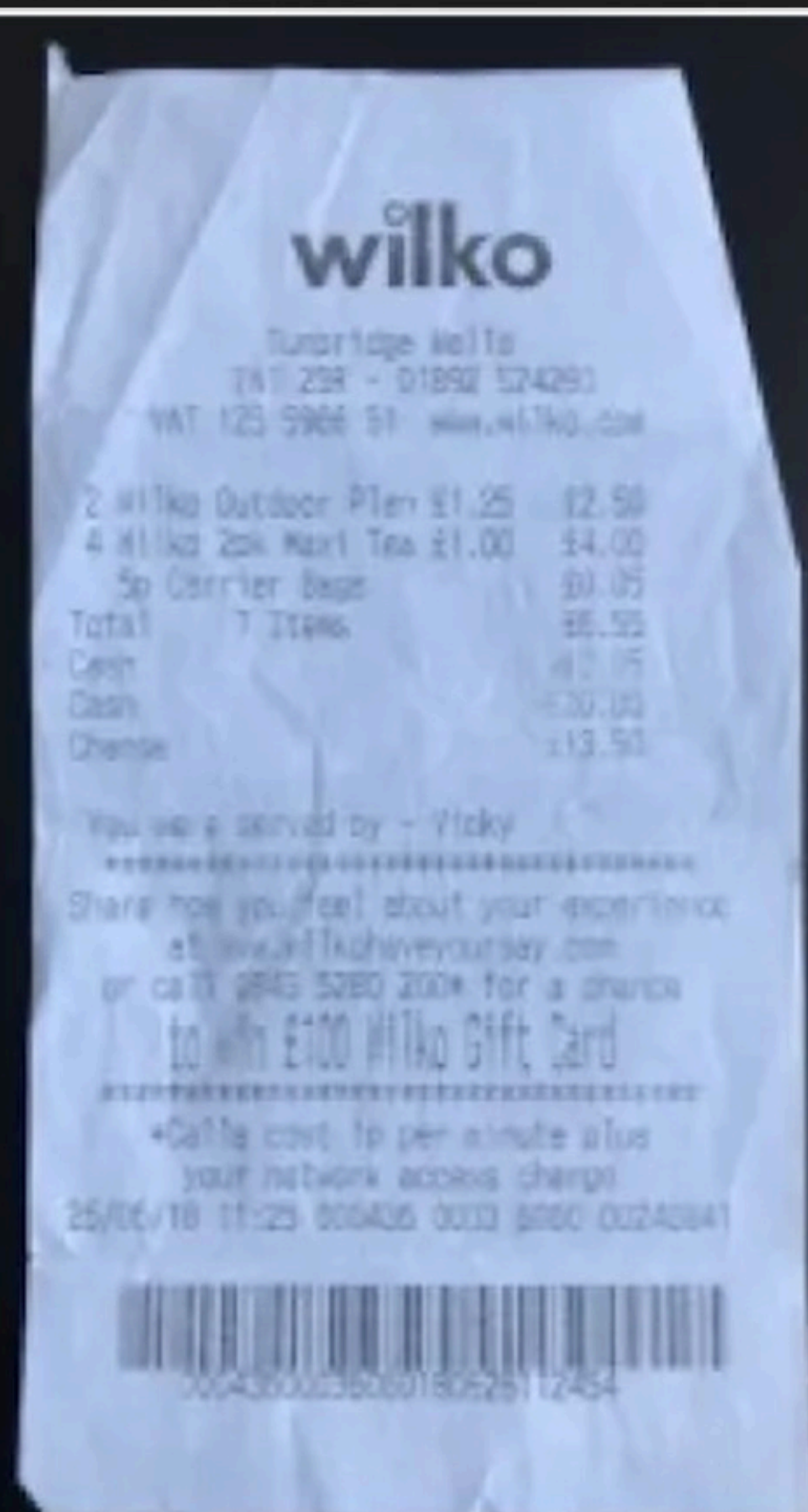
VAT Breakdown

Rate	Gross	Net	VAT
VAT @ 20%	69.99	58.33	11.66

Collection No: T367

Screwfix Direct Ltd Reg Office: Trade House Mead Avenue Yeovil BA22 8RT
Reg. No. 3006378 Registered in England Vat No: 232 5555 75
Telephone: 03330 112 112 24 hours a day, 7 days a week Online: www.screwfix.com

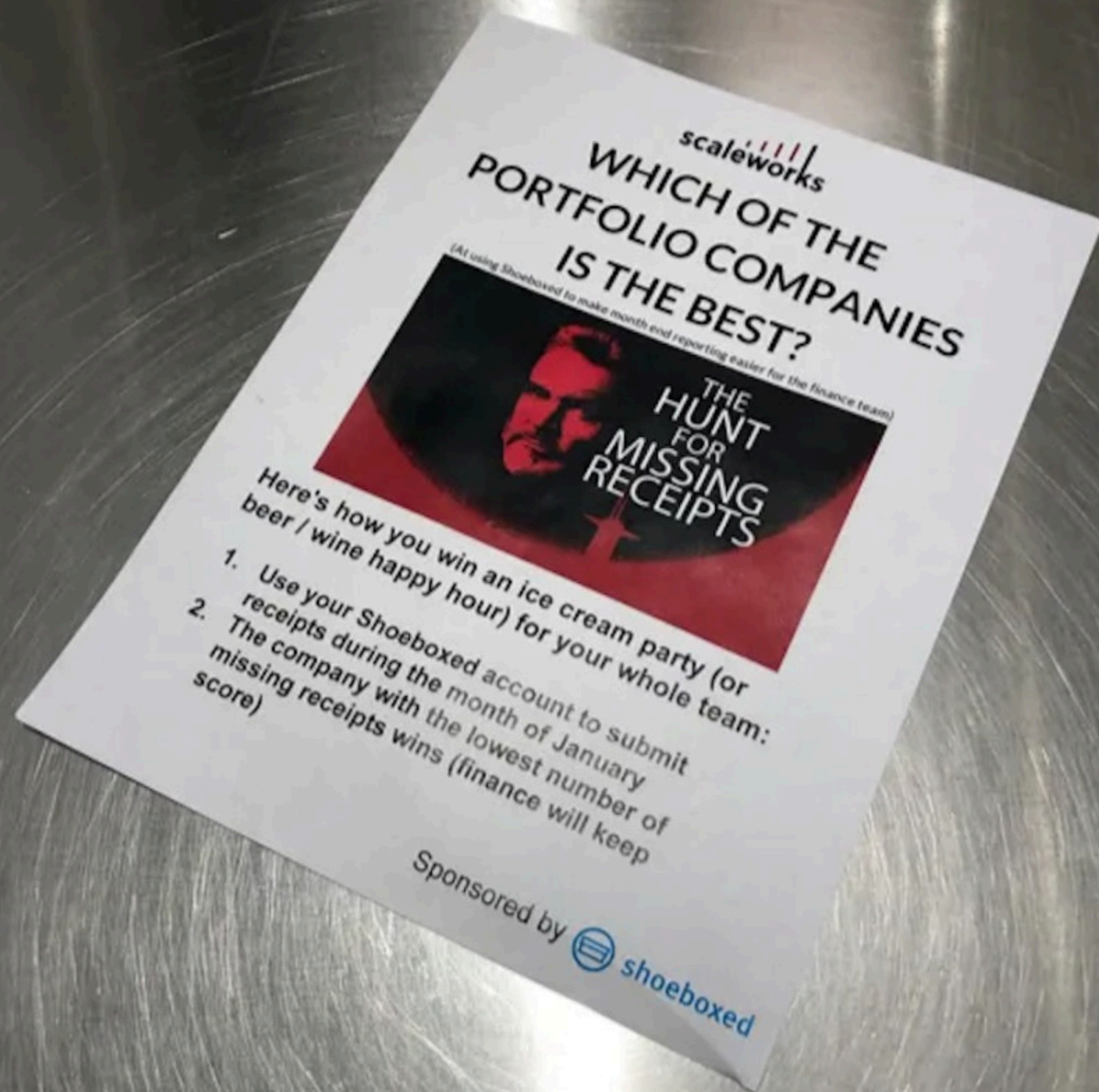
INPUT (480x640)



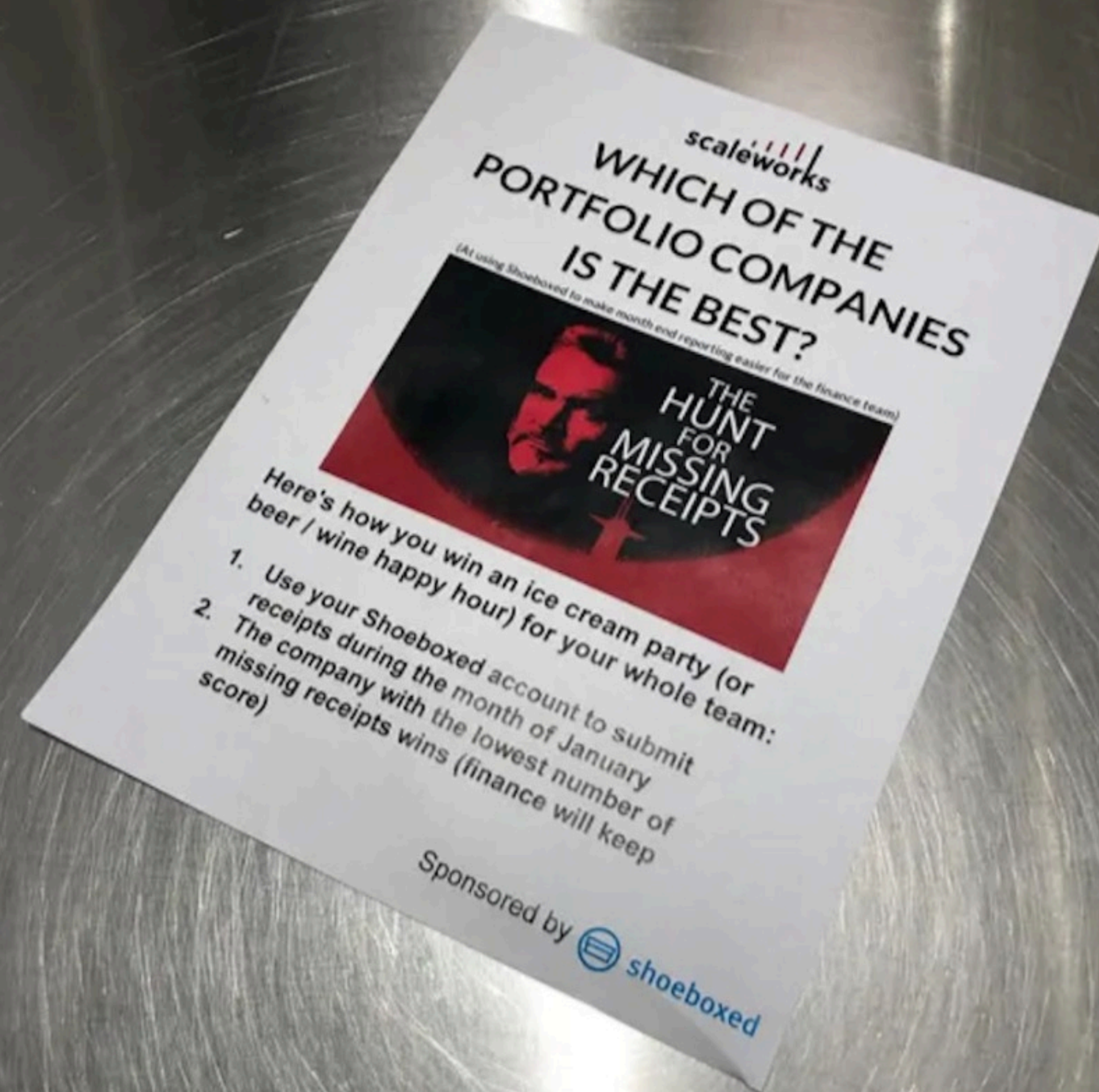
OUTPUT (322x614)



INPUT (1006x1010)



OUTPUT (1006x1010)



INPUT (886x1210)

Pumpkin
Oxford Station 12281371
VAT Registration No. 884 2579 78
Thankyou for your custom. Please call again.
.....

BILL	13992
------	-------

Latte 675ml	2.45
SUBTOTAL	2.45
CASH	2.50
CHANGE	0.05
SRNetSales	2.09
VAT 17.5%	0.36

CLK 4 / DRW 2
#001-004-2370-2680 21/12/2010 13:09-R

Select Service Partner
169 Euston Road London NW1 2AE
Comments welcome at: www.sspfeedback.com

OUTPUT (620x877)

Pumpkin
Oxford Station 12281371
VAT Registration No. 884 2579 78
Thankyou for your custom. Please call again.
.....

BILL	13992
------	-------

Latte 675ml	2.45
SUBTOTAL	2.45
CASH	2.50
CHANGE	0.05
SRNetSales	2.09
VAT 17.5%	0.36

CLK 4 / DRW 2
#001-004-2370-2680 21/12/2010 13:09-R

Select Service Partner
169 Euston Road London NW1 2AE
Comments welcome at: www.sspfeedback.com

Investigation of Long Short-Term Memorys to Temperature Prediction for Permanent Magnet Synchronous Motors

Oliver Willech, Wilfried Kierkgast, Joachim Böcker
Department of Power Electronics and Electrical Drive
Dortmund University
44226 Datteln, Germany
Email: willech@fbw.uni-dortmund.de

Abstract—Short-term prediction of temperature for permanent magnet synchronous motors (PMSMs) is critical in many applications and is a challenging task. In this paper, the long short-term memory (LSTM) neural network is investigated as a promising approach for temperature prediction. The LSTM is compared with other neural network models, such as recurrent neural networks (RNNs) and support vector machines (SVMs). The results show that the LSTM outperforms the other models in terms of prediction accuracy and robustness. The LSTM is able to capture the long-term dependencies in the temperature data, which is essential for accurate prediction. The results also show that the LSTM is robust to noise and outliers in the data. The proposed LSTM-based temperature prediction model is implemented in a real-time control system for a PMSM. The results show that the LSTM-based model is able to predict the temperature of the PMSM with high accuracy and low computational complexity.

1. INTRODUCTION

Permanent magnet synchronous motors (PMSMs) are preferred in many industrial applications (e.g. electrical vehicles, fans, pumps, etc.) due to their high efficiency and low maintenance. However, the temperature of the PMSM is a critical factor for its performance and reliability. High temperature can lead to demagnetization of the permanent magnets, which is irreversible and can cause significant damage to the motor. Therefore, it is essential to monitor and predict the temperature of the PMSM in real-time. In the past, several methods have been proposed for temperature prediction, such as using thermal models, empirical models, and machine learning. However, these methods often require a lot of data and are not very accurate. In this paper, we propose a new method for temperature prediction using Long Short-Term Memory (LSTM) neural networks. The LSTM is a type of recurrent neural network (RNN) that is designed to capture long-term dependencies in the data. It is able to learn the relationship between the input variables (e.g. current, voltage, speed) and the output variable (temperature) over a long period of time. The results show that the LSTM-based model is able to predict the temperature of the PMSM with high accuracy and low computational complexity.

(2). In both cases, temperatures are decreased by reducing the thermal properties of electric model parameters, e.g. the winding resistance or the PM flux linkage. However, as the temperature-dependent parameter sensitivity is very low even custom-tailored models can lead to significant prediction errors.

Direct thermal analysis can be used for online temperature prediction. Thermal capacity (CTH) and heat capacity (CH) models (CTH/CH) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136], [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [158], [159], [160], [161], [162], [163], [164], [165], [166], [167], [168], [169], [170], [171], [172], [173], [174], [175], [176], [177], [178], [179], [180], [181], [182], [183], [184], [185], [186], [187], [188], [189], [190], [191], [192], [193], [194], [195], [196], [197], [198], [199], [200], [201], [202], [203], [204], [205], [206], [207], [208], [209], [210], [211], [212], [213], [214], [215], [216], [217], [218], [219], [220], [221], [222], [223], [224], [225], [226], [227], [228], [229], [230], [231], [232], [233], [234], [235], [236], [237], [238], [239], [240], [241], [242], [243], [244], [245], [246], [247], [248], [249], [250], [251], [252], [253], [254], [255], [256], [257], [258], [259], [260], [261], [262], [263], [264], [265], [266], [267], [268], [269], [270], [271], [272], [273], [274], [275], [276], [277], [278], [279], [280], [281], [282], [283], [284], [285], [286], [287], [288], [289], [290], [291], [292], [293], [294], [295], [296], [297], [298], [299], [300], [301], [302], [303], [304], [305], [306], [307], [308], [309], [310], [311], [312], [313], [314], [315], [316], [317], [318], [319], [320], [321], [322], [323], [324], [325], [326], [327], [328], [329], [330], [331], [332], [333], [334], [335], [336], [337], [338], [339], [340], [341], [342], [343], [344], [345], [346], [347], [348], [349], [350], [351], [352], [353], [354], [355], [356], [357], [358], [359], [360], [361], [362], [363], [364], [365], [366], [367], [368], [369], [370], [371], [372], [373], [374], [375], [376], [377], [378], [379], [380], [381], [382], [383], [384], [385], [386], [387], [388], [389], [390], [391], [392], [393], [394], [395], [396], [397], [398], [399], [400], [401], [402], [403], [404], [405], [406], [407], [408], [409], [410], [411], [412], [413], [414], [415], [416], [417], [418], [419], [420], [421], [422], [423], [424], [425], [426], [427], [428], [429], [430], [431], [432], [433], [434], [435], [436], [437], [438], [439], [440], [441], [442], [443], [444], [445], [446], [447], [448], [449], [450], [451], [452], [453], [454], [455], [456], [457], [458], [459], [460], [461], [462], [463], [464], [465], [466], [467], [468], [469], [470], [471], [472], [473], [474], [475], [476], [477], [478], [479], [480], [481], [482], [483], [484], [485], [486], [487], [488], [489], [490], [491], [492], [493], [494], [495], [496], [497], [498], [499], [500], [501], [502], [503], [504], [505], [506], [507], [508], [509], [510], [511], [512], [513], [514], [515], [516], [517], [518], [519], [520], [521], [522], [523], [524], [525], [526], [527], [528], [529], [530], [531], [532], [533], [534], [535], [536], [537], [538], [539], [540], [541], [542], [543], [544], [545], [546], [547], [548], [549], [550], [551], [552], [553], [554], [555], [556], [557], [558], [559], [560], [561], [562], [563], [564], [565], [566], [567], [568], [569], [570], [571], [572], [573], [574], [575], [576], [577], [578], [579], [580], [581], [582], [583], [584], [585], [586], [587], [588], [589], [590], [591], [592], [593], [594], [595], [596], [597], [598], [599], [600], [601], [602], [603], [604], [605], [606], [607], [608], [609], [610], [611], [612], [613], [614], [615], [616], [617], [618], [619], [620], [621], [622], [623], [624], [625], [626], [627], [628], [629], [630], [631], [632], [633], [634], [635], [636], [637], [638], [639], [640], [641], [642], [643], [644], [645], [646], [647], [648], [649], [650], [651], [652], [653], [654], [655], [656], [657], [658], [659], [660], [661], [662], [663], [664], [665], [666], [667], [668], [669], [670], [671], [672], [673], [674], [675], [676], [677], [678], [679], [680], [681], [682], [683], [684], [685], [686], [687], [688], [689], [690], [691], [692], [693], [694], [695], [696], [697], [698], [699], [700], [701], [702], [703], [704], [705], [706], [707], [708], [709], [710], [711], [712], [713], [714], [715], [716], [717], [718], [719], [720], [721], [722], [723], [724], [725], [726], [727], [728], [729], [730], [731], [732], [733], [734], [735], [736], [737], [738], [739], [740], [741], [742], [743], [744], [745], [746], [747], [748], [749], [750], [751], [752], [753], [754], [755], [756], [757], [758], [759], [760], [761], [762], [763], [764], [765], [766], [767], [768], [769], [770], [771], [772], [773], [774], [775], [776], [777], [778], [779], [780], [781], [782], [783], [784], [785], [786], [787], [788], [789], [790], [791], [792], [793], [794], [795], [796], [797], [798], [799], [800], [801], [802], [803], [804], [805], [806], [807], [808], [809], [810], [811], [812], [813], [814], [815], [816], [817], [818], [819], [820], [821], [822], [823], [824], [825], [826], [827], [828], [829], [830], [831], [832], [833], [834], [835], [836], [837], [838], [839], [840], [841], [842], [843], [844], [845], [846], [847], [848], [849], [850], [851], [852], [853], [854], [855], [856], [857], [858], [859], [860], [861], [862], [863], [864], [865], [866], [867], [868], [869], [870], [871], [872], [873], [874], [875], [876], [877], [878], [879], [880], [881], [882], [883], [884], [885], [886], [887], [888], [889], [890], [891], [892], [893], [894], [895], [896], [897], [898], [899], [900], [901], [902], [903], [904], [905], [906], [907], [908], [909], [910], [911], [912], [913], [914], [915], [916], [917], [918], [919], [920], [921], [922], [923], [924], [925], [926], [927], [928], [929], [930], [931], [932], [933], [934], [935], [936], [937], [938], [939], [940], [941], [942], [943], [944], [945], [946], [947], [948], [949], [950], [951], [952], [953], [954], [955], [956], [957], [958], [959], [960], [961], [962], [963], [964], [965], [966], [967], [968], [969], [970], [971], [972], [973], [974], [975], [976], [977], [978], [979], [980], [981], [982], [983], [984], [985], [986], [987], [988], [989], [990], [991], [992], [993], [994], [995], [996], [997], [998], [999], [1000].

This contribution is a preliminary work for the above application. The idea is to use the additional degree of freedom using LSTM instead of abstract LSTM, trying to avoid the situation where beyond the state of the art, in literature only few publications can be found in this field, e.g. using backpropagation (BP) or recurrent neural

Investigation of Long Short-Term Memorys to Temperature Prediction for Permanent Magnet Synchronous Motors

Oliver Willech, Wilfried Kierkgast, Joachim Böcker
Department of Power Electronics and Electrical Drive
Dortmund University
44226 Datteln, Germany
Email: willech@fbw.uni-dortmund.de

Abstract—Short-term prediction of temperature for permanent magnet synchronous motors (PMSMs) is critical in many applications and is a challenging task. In this paper, the long short-term memory (LSTM) neural network is investigated as a promising approach for temperature prediction. The LSTM is compared with other neural network models, such as recurrent neural networks (RNNs) and support vector machines (SVMs). The results show that the LSTM outperforms the other models in terms of prediction accuracy and robustness. The LSTM is able to capture the long-term dependencies in the temperature data, which is essential for accurate prediction. The results also show that the LSTM is robust to noise and outliers in the data. The proposed LSTM-based temperature prediction model is implemented in a real-time control system for a PMSM. The results show that the LSTM-based model is able to predict the temperature of the PMSM with high accuracy and low computational complexity.

1. INTRODUCTION

Permanent magnet synchronous motors (PMSMs) are preferred in many industrial applications (e.g. electrical vehicles, fans, pumps, etc.) due to their high efficiency and low maintenance. However, the temperature of the PMSM is a critical factor for its performance and reliability. High temperature can lead to demagnetization of the permanent magnets, which is irreversible and can cause significant damage to the motor. Therefore, it is essential to monitor and predict the temperature of the PMSM in real-time. In the past, several methods have been proposed for temperature prediction, such as using thermal models, empirical models, and machine learning. However, these methods often require a lot of data and are not very accurate. In this paper, we propose a new method for temperature prediction using Long Short-Term Memory (LSTM) neural networks. The LSTM is a type of recurrent neural network (RNN) that is designed to capture long-term dependencies in the data. It is able to learn the relationship between the input variables (e.g. current, voltage, speed) and the output variable (temperature) over a long period of time. The results show that the LSTM-based model is able to predict the temperature of the PMSM with high accuracy and low computational complexity.

(2). In both cases, temperatures are decreased by reducing the thermal properties of electric model parameters, e.g. the winding resistance or the PM flux linkage. However, as the temperature-dependent parameter sensitivity is very low even custom-tailored models can lead to significant prediction errors.

Direct thermal analysis can be used for online temperature prediction. Thermal capacity (CTH) and heat capacity (CH) models (CTH/CH) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136], [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [158], [159], [160], [161], [162], [163], [164], [165], [166], [167], [168], [169], [170], [171], [172], [173], [174], [175], [176], [177], [178], [179], [180], [181], [182], [183], [184], [185], [186], [187], [188], [189], [190], [191], [192], [193], [194], [195], [196], [197], [198], [199], [200], [201], [202], [203], [204], [205], [206], [207], [208], [209], [210], [211], [212], [213], [214], [215], [216], [217], [218], [219], [220], [221], [222], [223], [224], [225], [226], [227], [228], [229], [230], [231], [232], [233], [234], [235], [236], [237], [238], [239], [240], [241], [242], [243], [244], [245], [246], [247], [248], [249], [250], [251], [252], [253], [254], [255], [256], [257], [258], [259], [260], [261], [262], [263], [264], [265], [266], [267], [268], [269], [270], [271], [272], [273], [274], [275], [276], [277], [278], [279], [280], [281], [282], [283], [284], [285], [286], [287], [288], [289], [290], [291], [292], [293], [294], [295], [296], [297], [298], [299], [300], [301], [302], [303], [304], [305], [306], [307], [308], [309], [310], [311], [312], [313], [314], [315], [316], [317], [318], [319], [320], [321], [322], [323], [324], [325], [326], [327], [328], [329], [330], [331], [332], [333], [334], [335], [336], [337], [338], [339], [340], [341], [342], [343], [344], [345], [346], [347], [348], [349], [350], [351], [352], [353], [354], [355], [356], [357], [358], [359], [360], [361], [362], [363], [364], [365], [366], [367], [368], [369], [370], [371], [372], [373], [374], [375], [376], [377], [378], [379], [380], [381], [382], [383], [384], [385], [386], [387], [388], [389], [390], [391], [392], [393], [394], [395], [396], [397], [398], [399], [400], [401], [402], [403], [404], [405], [406], [407], [408], [409], [410], [411], [412], [413], [414], [415], [416], [417], [418], [419], [420], [421], [422], [423], [424], [425], [426], [427], [428], [429], [430], [431], [432], [433], [434], [435], [436], [437], [438], [439], [440], [441], [442], [443], [444], [445], [446], [447], [448], [449], [450], [451], [452], [453], [454], [455], [456], [457], [458], [459], [460], [461], [462], [463], [464], [465], [466], [467], [468], [469], [470], [471], [472], [473], [474], [475], [476], [477], [478], [479], [480], [481], [482], [483], [484], [485], [486], [487], [488], [489], [490], [491], [492], [493], [494], [495], [496], [497], [498], [499], [500], [501], [502], [503], [504], [505], [506], [507], [508], [509], [510], [511], [512], [513], [514], [515], [516], [517], [518], [519], [520], [521], [522], [523], [524], [525], [526], [527], [528], [529], [530], [531], [532], [533], [534], [535], [536], [537], [538], [539], [540], [541], [542], [543], [544], [545], [546], [547], [548], [549], [550], [551], [552], [553], [554], [555], [556], [557], [558], [559], [560], [561], [562], [563], [564], [565], [566], [567], [568], [569], [570], [571], [572], [573], [574], [575], [576], [577], [578], [579], [580], [581], [582], [583], [584], [585], [586], [587], [588], [589], [590], [591], [592], [593], [594], [595], [596], [597], [598], [599], [600], [601], [602], [603], [604], [605], [606], [607], [608], [609], [610], [611], [612], [613], [614], [615], [616], [617], [618], [619], [620], [621], [622], [623], [624], [625], [626], [627], [628], [629], [630], [631], [632], [633], [634], [635], [636], [637], [638], [639], [640], [641], [642], [643], [644], [645], [646], [647], [648], [649], [650], [651], [652], [653], [654], [655], [656], [657], [658], [659], [660], [661], [662], [663], [664], [665], [666], [667], [668], [669], [670], [671], [672], [673], [674], [675], [676], [677], [678], [679], [680], [681], [682], [683], [684], [685], [686], [687], [688], [689], [690], [691], [692], [693], [694], [695], [696], [697], [698], [699], [700], [701], [702], [703], [704], [705], [706], [707], [708], [709], [710], [711], [712], [713], [714], [715], [716], [717], [718], [719], [720], [721], [722], [723], [724], [725], [726], [727], [728], [729], [730], [731], [732], [733], [734], [735], [736], [737], [738], [739], [740], [741], [742], [743], [744], [745], [746], [747], [748], [749], [750], [751], [752], [753], [754], [755], [756], [757], [758], [759], [760], [761], [762], [763], [764], [765], [766], [767], [768], [769], [770], [771], [772], [773], [774], [775], [776], [777], [778], [779], [780], [781], [782], [783], [784], [785], [786], [787], [788], [789], [790], [791], [792], [793], [794], [795], [796], [797], [798], [799], [800], [801], [802], [803], [804], [805], [806], [807], [808], [809], [810], [811], [812], [813], [814], [815], [816], [817], [818], [819], [820], [821], [822], [823], [824], [825], [826], [827], [828], [829], [830], [831], [832], [833], [834], [835], [836], [837], [838], [839], [840], [841], [842], [843], [844], [845], [846], [847], [848], [849], [850], [851], [852], [853], [854], [855], [856], [857], [858], [859], [860], [861], [862], [863], [864], [865], [866], [867], [868], [869], [870], [871], [872], [873], [874], [875], [876], [877], [878], [879], [880], [881], [882], [883], [884], [885], [886], [887], [888], [889], [890], [891], [892], [893], [894], [895], [896], [897], [898], [899], [900], [901], [902], [903], [904], [905], [906], [907], [908], [909], [910], [911], [912], [913], [914], [915], [916], [917], [918], [919], [920], [921], [922], [923], [924], [925], [926], [927], [928], [929], [930], [931], [932], [933], [934], [935], [936], [937], [938], [939], [940], [941], [942], [943], [944], [945], [946], [947], [948], [949], [950], [951], [952], [953], [954], [955], [956], [957], [958], [959], [960], [961], [962], [963], [964], [965], [966], [967], [968], [969], [970], [971], [972], [973], [974], [975], [976], [977], [978], [979], [980], [981], [982], [983], [984], [985], [986], [987], [988], [989], [990], [991], [992], [993], [994], [995], [996], [997], [998], [999], [1000].

This contribution is a preliminary work for the above application. The idea is to use the additional degree of freedom using LSTM instead of abstract LSTM, trying to avoid the situation where beyond the state of the art, in literature only few publications can be found in this field, e.g. using backpropagation (BP) or recurrent neural

